

TEACHER NOTES — DO NOT DISTRIBUTE

Class 4 — Composition, Inverses, Applications & Multivariable Functions

Math Summer Camp

Talk about the domain/codomain fix: we can **not** leave elements out of the domain.

Comparing the Infinite — Teacher Talking Points

1. How to compare infinite sets

Open on the gap. For finite sets, “same size” is easy: count both, compare the numbers. But you can never finish counting $\mathbb{N}$ — counting is useless for infinite sets. We need a different tool, and we have secretly used it all homework: **pairing**.

Shepherd analogy: match each sheep to one pen-spot; if none are left standing and no spot is empty, the flocks are equal — and you never counted.

The precision point (theorem vs definition). For *finite* sets, “a bijection exists $\iff$ equal count” is a **theorem** (prove it by pairing / Pigeonhole) — and notice the pairing never uses a number. That number-free test is exactly what still works once counting dies, so for *infinite* sets we **adopt** “same size = a bijection exists” as the **definition** (Cantor’s move). Do not try to prove the infinite version — that is a category error. Prove it where you can (finite), see it needs no counting, and promote it to the definition everywhere else.

Theorem. For finite sets A and B , there is a bijection $A \rightarrow B$ if and only if A and B have the same number of elements ($|A| = |B|$).

What “count” means. Saying $|A| = n$ is *itself* defined by a bijection: A has n elements means there is a bijection $\alpha: A \rightarrow \{1, 2, \dots, n\}$ (a labeling of the elements 1 through n). So “count” is just “can be paired with the standard set $\{1, \dots, n\}$.” The theorem then rests on two facts:

- (i) a composition of bijections is a bijection, and the inverse of a bijection is a bijection;
- (ii) **Pigeonhole:** if $m > n$, no function $\{1, \dots, m\} \rightarrow \{1, \dots, n\}$ can be injective.

Proof.

($\Leftarrow$) Suppose $|A| = |B| = n$. Then there are bijections $\alpha: A \rightarrow \{1, \dots, n\}$ and $\beta: B \rightarrow \{1, \dots, n\}$. By (i), the composite $\beta^{-1} \circ \alpha: A \rightarrow B$ is a bijection. So a bijection $A \rightarrow B$ exists.

($\Rightarrow$) Suppose $f: A \rightarrow B$ is a bijection, with $|A| = m$ and $|B| = n$ via labelings $\alpha: A \rightarrow \{1, \dots, m\}$ and $\beta: B \rightarrow \{1, \dots, n\}$. Then

$$\beta \circ f \circ \alpha^{-1}: \{1, \dots, m\} \rightarrow \{1, \dots, n\}$$

is a bijection (a composition of bijections). It remains to show such a bijection forces $m = n$:

- it is injective, so by Pigeonhole (ii) we cannot have $m > n$; hence $m \leq n$;
- its inverse $\{1, \dots, n\} \rightarrow \{1, \dots, m\}$ is also injective, so likewise $n \leq m$.

Therefore $m = n$, i.e. $|A| = |B|$. $\square$

Remark. The whole argument is pure pairing — it never counts anything by hand. That is exactly why the criterion “same size = a bijection exists” survives into the infinite case, where Pigeonhole (and the uniqueness of n) no longer applies.

Definition. Two sets have the **same size** (same *cardinality*) if there is a **bijection** between them — a perfect pairing, nothing doubled up, nothing left over. It never counts anything; it only asks whether every element of one can be matched with exactly one of the other.

3. The proper-subset paradox (resolving the homework itch)

Start from their result. $f(n) = 2n$ is a bijection from $\mathbb{N}$ onto the evens E , so $\mathbb{N}$ and E are the **same size** — even though E throws away every odd number.

The resolution. That collides with the planted imprecision, “a set always has strictly more elements than any of its proper parts.” It is **true for finite sets only**; for infinite sets it is false, and $\mathbb{N} \leftrightarrow E$ is the counterexample.

Reframe it as a feature, not a bug. Being able to match one-to-one with a proper part of itself is precisely what **defines** an infinite set (Dedekind). Finite sets cannot (that is the Finite Trap from the homework); infinite sets always can.

Honesty note so they do not over-generalize: the rule “if $|A| > |B|$ then no injection” still holds even for infinite sets — it is the definition of $>$. What breaks is the finite intuition “proper subset $\Rightarrow$ smaller.” Takeaway: you cannot judge the size of an infinite set by containment or by counting — only by building a bijection.

2. Cantor’s argument — countable vs. uncountable

Definition. A set is **countable** if you can arrange all of it in one endless list $a_1, a_2, a_3, \dots$ with no repeats and nothing missing. (The list *is* a bijection with $\mathbb{N}$.)

- **$\mathbb{Z}$ is countable:** list it $0, 1, -1, 2, -2, 3, -3, \dots$ — every integer appears once.
- **$\mathbb{Q}$ is countable (the surprise):** put every fraction in a grid (numerator $\times$ denominator) and sweep diagonally, skipping repeats — every fraction is eventually reached. So the rationals, which *feel* far denser, are the same size as $\mathbb{N}$.
- *Hilbert’s Hotel:* a full hotel still fits a newcomer by shifting room $n \rightarrow n + 1$. “Always room for one more” is the signature of a countable infinity.

The reveal — $\mathbb{R}$ is uncountable (the diagonal argument). Suppose someone hands you a *complete* list of all reals in $[0, 1]$ as decimals. Build a brand-new number by walking down the diagonal and *changing* each digit: make the n th digit of the new number differ from the n th digit of the n th listed number (e.g. use 5, or 6 if it is already 5). This new number differs from every listed number in at least one place, so it cannot be on the list. No list can hold all the reals, so $\mathbb{R}$ is a **strictly bigger** infinity — not all infinities are equal; there is a whole ladder of them. (*The 5/6 digit trick dodges the $0.4999\dots = 0.5$ ambiguity.*)

Part 1 — Building and Undoing Functions

Teaching notes. Socratic opener: “You needed 2 wool per block and $10x$ blocks — did you do that in one leap or two steps?” Surface the hidden composition. Order matters: $g \circ f \neq f \circ g$ in general — reuse the homework numbers $f(x) = x + 1$, $g(x) = 2x$: $(g \circ f)(3) = 8$ but $(f \circ g)(3) = 7$. Curveball comes after the example.

Composition

DEFINITION. Given $f: A \rightarrow B$ and $g: B \rightarrow C$, the **composition** $g \circ f: A \rightarrow C$ is defined by

$$(g \circ f)(x) = g(f(x)).$$

Apply the inside function first. Read it “ g of f .” (The outputs of f must lie in the domain of g , or the chaining is undefined.)

INTUITION. Two machines in a line: the first machine’s output is fed straight into the second.

EXAMPLE (the one they already did without noticing). A room is 10 blocks long and x blocks wide.

$$c(x) = 10x \quad (\text{carpet blocks needed}), \quad w(b) = 2b \quad (\text{wool for } b \text{ blocks}).$$

Total wool as a function of width is the composition

$$(w \circ c)(x) = w(c(x)) = w(10x) = 2(10x) = 20x.$$

The “20x” they wrote in one step IS $w \circ c$ — they composed two machines (blocks-per-width, then wool-per-block) without naming it.

Curveball — ask for the other order, $(c \circ w)(x)$. The legitimate composition is $w \circ c$: width $\rightarrow$ blocks $\rightarrow$ wool, with $(w \circ c)(x) = w(10x) = 20x$. Each machine’s output type matches the next machine’s input: c hands out “blocks,” and w accepts “blocks” — the hose connects. Now reverse it: feed x into w first, then into c :

$$(c \circ w)(x) = c(w(x)) = c(2x) = 10 \cdot (2x) = 20x.$$

The trap: the algebra gives $20x$ — the *same number* as $w \circ c$ — so a student watching only the arithmetic concludes “same answer, order does not matter, both fine.” But $c \circ w$ is **meaningless**: $w(x)$ outputs *wool*, while c expects a *width*. So $c \circ w$ shoves a wool amount into a slot that only accepts widths — “the width of a room, measured in wool” is nonsense.

Formally: for $g \circ f$ to even be defined, the inner function’s outputs must land in the outer function’s domain — $\text{range}(f) \subseteq \text{domain}(g)$. Here $\text{range}(w)$ is wool amounts and $\text{domain}(c)$ is widths, and wool amounts are not widths, so the composition does not exist — no matter what the algebra spits out. (*The numbers coincide only because c and w are both “multiply by a constant”; do not let them generalize that composition is commutative — contrast with $f(x) = x + 1$, $g(x) = 2x$, where $g \circ f \neq f \circ g$.*)

Inverse

DEFINITION. If $f: A \rightarrow B$ is a **bijection**, its **inverse** $f^{-1}: B \rightarrow A$ sends each output back to the input it came from:

$$f^{-1}(y) = x \quad \text{exactly when} \quad f(x) = y.$$

Only bijections have inverses.

INTUITION. The “undo” machine — run the function backwards.

EXAMPLE (bedroom). Single-room wool $f(x) = 20x$. Given a wool count W , recover the width: $f^{-1}(W) = W/20$. The original did “ $\times 10$ then $\times 2$ ”; the inverse undoes in *reverse order*: “ $\div 2$ then $\div 10$ ” ($= \div 20$). For the three-bedroom $f(w) = 20w + 280$: the inverse subtracts 280, then divides by 20.

Socks-and-shoes: to undo “socks then shoes,” you take off shoes then socks — the inverse of a composition reverses the order: $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Why only bijections (tie to the homework planted imprecision). If f is not injective, two inputs share an output and the undo machine cannot choose; if f is not surjective, some

codomain value is never produced and the undo machine has no input for it. This is exactly why “every function can be undone, so every function has an inverse” (HW Looking Ahead) was **false**. Connect to the HW “Answer this” question.

Part 2 — Functions in the Real World

*Injective / surjective / bijective were defined in Class 3. Here we interpret them, and ask what **proving** each one actually buys you.*

Injective in the wild

MEANING. Different inputs give different outputs — no two inputs collide; each input has a unique fingerprint.

WHAT PROVING IT GUARANTEES. You can always tell inputs apart by their output; the output is a reliable identifier; distinct things stay distinct.

EXAMPLES. Unique student IDs; fingerprints; a price list where each item has its own price (so a price names the item). *Curveball non-example:* “assign each person their birth month” is NOT injective (12 outputs, many collisions) — you cannot identify a person from their birth month.

Surjective in the wild

MEANING. Every element of the codomain is hit — every target is achievable; full coverage.

WHAT PROVING IT GUARANTEES. Nothing in the declared target is unreachable.

EXAMPLES. A vending machine where every product can be dispensed by some code; a grading rule that can output every grade A–F. *Non-example:* the wool function with codomain $\mathbb{Z}^+$ is NOT surjective (only multiples of 20 from 300 up) — most wool counts are impossible totals.

Bijective in the wild

MEANING. A perfect one-to-one pairing (injective *and* surjective) — a reversible code.

WHAT PROVING IT GUARANTEES. The process can be perfectly undone: it **has an inverse**. You can encode AND decode with no loss and no ambiguity. (f bijective $\iff f$ invertible.)

EXAMPLES. A Caesar cipher / ROT13 (each message $\leftrightarrow$ exactly one ciphertext, decodable); a seat $\leftrightarrow$ ticket assignment.

Application problems (use for the EXAMPLE slots).

- A locker system gives each of 30 students a distinct 4-digit code. Injective? **Yes** (distinct). Surjective onto all 10000 codes? **No**. Bijective? **No**. What does injectivity buy you here? No two students share a code — a code identifies its student.
- A machine shifts every letter forward by 3 (Caesar cipher). Bijective? **Yes** — undo by shifting back 3. Why does that matter? It is exactly what makes the message decodable.

Problem A — build one function. A taco truck charges a flat \$4 to park on your street, plus \$2.50 for each taco you order. Write a function f for the total cost (in dollars) of ordering t tacos. Describe it fully: give the definition ($f: A \rightarrow B$, naming the domain and codomain) and the rule.

Solution. $f: \mathbb{N} \rightarrow \mathbb{R}$ (domain = whole numbers of tacos $\{0, 1, 2, \dots\}$ — you cannot order half a taco; codomain = dollars). Rule: $f(t) = 4 + 2.5t$. *Talking point: the rule looks like a straight line, but because the domain is discrete the graph is just isolated dots, and the range is the discrete set $\{4, 6.5, 9, \dots\}$ — a clean domain/codomain check.*

Problem B — build two functions, then compose. In Minecraft, 1 log crafts into 4 planks, and each plank crafts into 2 sticks. You start with L logs. **(a)** Write a function p for the planks from L logs, and a function s for the sticks from P planks — define each ($\rightarrow$ and rule). **(b)** Use composition to write a single function for the sticks from L logs. Is it $s \circ p$ or $p \circ s$, and why?

Solution. (a) $p: \mathbb{N} \rightarrow \mathbb{N}$, $p(L) = 4L$ (planks from logs); $s: \mathbb{N} \rightarrow \mathbb{N}$, $s(P) = 2P$ (sticks from planks). (b) Sticks-from-logs is $s \circ p$: $(s \circ p)(L) = s(4L) = 8L$. It must be $s \circ p$, not $p \circ s$ — p outputs *planks*, which is exactly what s takes in; doing $p \circ s$ would feed *sticks* into the planks-machine (the same type-mismatch as the $c \circ w$ curveball).

Part 3 — Multivariable Functions

Multivariable function (two inputs)

DEFINITION. A function $f: A \times B \rightarrow C$ takes an **ordered pair** (x, y) as input and returns one output, written $f(x, y)$. The domain is a set of ordered pairs — a subset of a Cartesian product (callback to the sets unit).

INTUITION. A machine with **two input slots**. The bedroom with two undecided widths: the total wool depends on both.

EXAMPLE. Two rooms, each 10 long, widths w_1, w_2 :

$$f(w_1, w_2) = 20w_1 + 20w_2, \quad f(3, 5) = 20(3) + 20(5) = 160.$$

Domain = $\mathbb{Z}^+ \times \mathbb{Z}^+$ (pairs of positive integers).

Domain, codomain, range (two inputs)

Domain = the allowed input *pairs* (here $\mathbb{Z}^+ \times \mathbb{Z}^+$). **Codomain** = the declared outputs (e.g. $\mathbb{Z}^+$). **Range** = the totals actually achievable (multiples of 20 that are ≥ 40). Same trio as before — the only new thing is that the domain lives inside a Cartesian product.

Graphing a two-input function (a 3D surface)

INTUITION. The input pair (x, y) is a point on a flat floor (the xy -plane); the output is the **height** above that point. The graph is a **surface** (a landscape), not a curve. *Analogy: temperature at each spot on a map, or elevation on a topographic map.*

EXAMPLE. $f(x, y) = 20x + 20y$ is a tilted flat plane: pick a few floor points (x, y) , plot the height, see it rise as you move in either direction.

Injective / surjective / bijective with pairs

Teaching note. Let them *discover* $f(1, 2) = f(2, 1)$ themselves — it lands the whole arc (composition $\rightarrow$ multivariable $\rightarrow$ injectivity $\rightarrow$ inverses) in one concrete example. This caps the Functions milestone.

INTUITION. “Different inputs” now means different *pairs*.

EXAMPLE (the key curveball). Is $f(w_1, w_2) = 20w_1 + 20w_2$ injective? **No.**

$$f(1, 2) = 60 = f(2, 1), \quad \text{but} \quad (1, 2) \neq (2, 1).$$

Trading width between the two rooms gives the same total — many different pairs share an output.

CONSEQUENCE (ties back to inverses). Because it is not injective, you *cannot* recover both widths from the total wool alone — there is no inverse. Two-input functions are usually not injective, which is exactly why you generally cannot “undo” them to a single input pair.

A multivariable function that IS injective (contrast the bedroom). The bedroom function $f(w_1, w_2) = 20w_1 + 20w_2$ was *not* injective — it crushes a pair down to one number. Here is a three-in, three-out function that *is* injective:

$$f(x, y, z) = (x + y, x - y, z), \quad f: \mathbb{R}^3 \rightarrow \mathbb{R}^3.$$

Why it works. Injective means different input points must give different output points. Suppose instead two inputs share an output, $f(x_1, y_1, z_1) = f(x_2, y_2, z_2)$. Matching the three coordinates:

$$x_1 + y_1 = x_2 + y_2,$$

$$x_1 - y_1 = x_2 - y_2,$$

$$z_1 = z_2.$$

Adding the first two equations: $2x_1 = 2x_2$, so $x_1 = x_2$. Subtracting the second from the first: $2y_1 = 2y_2$, so $y_1 = y_2$. And $z_1 = z_2$ directly. Hence $(x_1, y_1, z_1) = (x_2, y_2, z_2)$ — the two inputs were the same after all. Therefore f is injective. $\square$

Teaching note. Unlike the bedroom function, this one loses no information: you can recover (x, y, z) from the outputs (add and subtract the first two, read off the third), so it is actually a bijection and *has an inverse*. It is also a **linear map** — a clean bridge into next week’s Linear Algebra.